

ABSTRACT

MODULAR KNOWLEDGE-BASE CONSTRUCTION FOR MULTI-DOMAIN OMNI-RAG: EVALUATING CHUNKERS, INDEXERS, AND A LEGAL KNOWLEDGE GRAPH

By:

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Retrieval-Augmented Generation (RAG) systems must turn heterogeneous documents into searchable information units before a language model can produce an answer. This paper presents a modular approach based on a canonical representation and a common *plugin* interface for knowledge-base construction in OMNI-RAG. The evaluation does not examine plugin internals; instead, it compares *chunker*, *indexer*, and *knowledge-graph* strategies while changing one component at a time.

The experiments use nine Netgear *User Manuals* comprising 1,400 pages and 200 bilingual questions, together with a legal corpus containing 256 documents and 25 verified questions. At $K = 30$, the *structure-aware hybrid* configuration reaches a recall of 0.740 on the User Manual corpus, while *structure-aware dense* reaches 0.785. On the legal corpus, *legal-structure dense* reaches a recall of 0.6854, whereas *structure-aware hybrid* achieves the highest MRR at 0.5439. A knowledge-graph traversal to depth two reaches a neighborhood recall of 1.0000 with the *sliding* configuration.

The results indicate that preserving document structure and matching semantics improve context coverage, but the best configuration depends on the domain and the metric being prioritized. They also show that a *plugin* abstraction can serve as both a strategy-exchange boundary and an experimental control mechanism without changing the core orchestrator.

Keywords: OMNI-RAG, Retrieval-Augmented Generation, knowledge-base construction, chunking, indexer, knowledge graph, plugin architecture.